



Play Something

Netflix has rolled out its 'Play Something' shuffle feature for TVs and living room devices, globally. <http://digit.in/may21-43>



Truecaller Covid-19 directory

Truecaller has launched a COVID-19 Healthcare directory to help find details of hospitals nearby. <http://digit.in/may21-44>

Active Noise Cancellation 101

Learn about the different types of ANC and how they work

Manish Rajesh | manish@digit.in

N

umerous headphones today come with ANC technology. ANC or Active Noise

Cancelling tech was something that was reserved for the most expensive of headphones until just a few years ago. Today, you'll find even budget offerings with ANC. However, not all implementations of ANC are the same. There are different types of design and technology that come into play with the various different implementations of ANC technology. Each of these implementations affect the overall quality of the ANC that a device can produce. In this feature, we'll be taking a look at the different types of ANC and how they work.

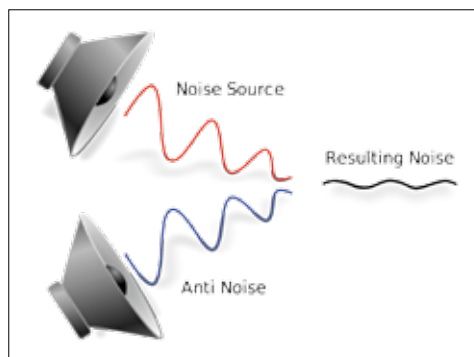
WHAT IS ANC

Active Noise Cancelling is based on something called the principle of wave phase cancellation. In simple words, sound is created in waves and what Active Noise Cancelling does, is record the sounds in your background and invert the sound signal, or the waves that the background sound is creating. This creates something called "anti-noise", which is played along with whatever it is you're listening to. This "anti-noise" signal, which is the inverted sound signal of the background sounds,

cancels out all the noise it's picking up from around you. It's fairly simple in theory, and the idea dates all the way back to the 1930s, but the execution is far from easy.

There are a few factors that come into play in order for ANC to work. First, it would need to flawlessly capture background sound. We're not at a stage where microphones can do that yet, and chances are, we won't be seeing 100 per cent noise cancellation for a long while. However, what we can achieve is still fairly impressive, and if you've tried a good pair of

ANC headphones yourself, then you can vouch for that. The second factor that needs to be considered is the form factor, open-back headphones, closed-back headphones, earphones, all of these affect how sound is captured and affects the quality of ANC. Of course, form factor also affects how much background sound can actually get through the seal, for instance, if your earphones



$$\text{Noise} + \text{Anti-noise} = \text{ANC}$$